

# Matthias Rottner

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## Current Position

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|---------|---------------------------------------------------------------------------------|
| 07/2024 | Economist, Bank for International Settlements, Monetary and Economic Department |
| 05/2021 | Economist, Deutsche Bundesbank, Research Centre (on leave since 07/2024)        |

## Education

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|-------------------|-------------------------------------------------------------------------------------------|
| 09/2016 - 05/2021 | Ph.D. in Economics, European University Institute<br>Advisors: Evi Pappa, Leonardo Melosi |
| 09/2016 - 08/2017 | MRes in Economics, European University Institute                                          |
| 09/2014 - 08/2016 | MSc in Economics, University of Copenhagen                                                |
| 04/2011 - 03/2014 | BA in Economics, University of Erlangen-Nürnberg                                          |

## Research and Professional Experience

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|-------------------|------------------------------------------------------|
| 02/2023 - 03/2023 | Visiting Researcher, Bank of Estonia                 |
| 09/2020 - 04/2021 | Consultant, European Central Bank                    |
| 09/2019 - 08/2020 | PhD Traineeship, European Central Bank               |
| 01/2019 - 06/2019 | Visiting Researcher, Federal Reserve Bank of Chicago |
| 01/2019 - 06/2019 | Visiting Researcher, Northwestern University         |
| 08/2018 - 12/2018 | Internship, Deutsche Bundesbank                      |
| Summer 2015,16,17 | Internship, Bank of Estonia                          |
| 04/2014 - 06/2014 | Internship, Kiel Institute for the World Economy     |

## Publications

**Geopolitical Risk Perceptions**, with Y. Bondarenko, V. Lewis and Y. Schüler, Journal of International Economics, 2024, Volume 152. - [Link](#)

**Reversal Interest Rate and Macroprudential Policy**, with M. Darracq Pariès and C. Kok, European Economic Review, 2023, Volume 159. - [Link](#)

**Financial Crises and Shadow Banks: A Quantitative Analysis**, Journal of Monetary Economics, 2023, Volume 139, p. 74-92. - [Link](#)

**Pandemic Recessions and Contact Tracing** with L. Melosi, Journal of the European Economic Association, 2023, Volume 21(6), p. 2485-2517. - [Link](#)

**Hitting The Elusive Inflation Target**, with F. Bianchi and L. Melosi, Journal of Monetary Economics, 2021, Volume 124, p. 107-122. - [Link](#)

## Working Papers

**Estimating Nonlinear Heterogeneous Agent Models with Neural Networks**, with H. Kase and L. Melosi, **R&R Econometrica** (2nd round) - [Link](#)

**CBDC and Banks: Disintermediating fast and slow**, with R. Bidder and T. Jackson, **R&R Journal of Financial Economics** - [Link](#)

**Learning Monetary Policy Strategies at the Effective Lower Bound with Sudden Surprises**, with S. Krane and L. Melosi, **R&R Review of Economic Dynamics** - [Link](#)

**The Perils of Narrowing Fiscal Spaces**, with H. Kase, L. Melosi and S. Rast - [Link](#)

**Geopolitical Risk in the Euro Area: Measurement and Transmission**, with Y. Bondarenko, V. Lewis, N. Kang and Y. Schüler - [Link](#)

**The Macroeconomics of Stablecoins**, with B. Hofmann and M. Kaldorf - [Link](#)

**Generative Economic Modeling**, with H. Kase and F. Stohler - [Link](#)

**Monetary Policy and Earnings Inequality: Inflation Dependencies**, with J. Meriküll - [Link](#)

**Climate Minsky Moments and Endogenous Financial Crises**, with M. Kaldorf - [Link](#)

## Work in Progress

**Economic Narratives and Realities of Geopolitical Risk**, with S. Arndt, Y. Bondarenko, V. Lewis and Y. Schueler

**Nonlinear Phillips Curve and Inflation Risk**, with H. Kase, L. Melosi and S. Rast

## Other Publications

**The impact of geopolitical risk on the euro area economy: Past experience and future prospects** with Y. Bondarenko, N. Kang, V. Lewis and Y. Schüler, VoxEU Article, 24 June 2025 - [Link](#)

**Evolving approaches to monetary policy communication in the face of uncertainty: fan charts, scenarios and guidance** with S. Bell, M. Chavaz, B. Hofmann, D. Rees, BIS Quarterly Review, March 2026 - [Link](#)

**The BIS multisector model: a multi-country environment for macroeconomic analysis** with M. Burgert, G. Cornelli, B. Erik, B. Mojon, D. Rees, H. Zhao, BIS Working Papers 1297, October 2025 - [Link](#)

**A multi-sector assessment of the macroeconomic effects of tariffs** with M. Burgert, B. Mojon, D. Rees, H. Zhao, BIS Quarterly Review, September 2025 - [Link](#)

**Macroeconomic effects of carbon-intensive energy price changes: A model comparison** with M. Burgert, M. Darracq Pariès, L. Durand, M. Gonzalés, R. Priftis, O. Röhe, E. Silgado-Gómez, N. Stähler, J. Varga, NGFS (Network for Greening the Financial System) Occasional Paper, September 2025 - [Link](#)

**Geopolitical risk in the euro area** with Y. Bondarenko, V. Lewis and Y. Schüler, VoxEU Article, 24 June 2025 - [Link](#)

**Climate Minsky moments: Why they may matter less than expected** with M. Kaldorf; VoxEU Article, 5 Feb 2025 - [Link](#)

**Commodity prices and monetary policy: old and new challenges** with F. Avalos, R. Banerjee, M. Burgert, B. Hofmann and C. Manea; BIS Bulletin, No. 96, 8 January 2025 - [Link](#)

**The digital euro can strengthen financial stability, with limits** with R. Bidder and T. Jackson; VoxEU Article, 3 July 2024 - [Link](#)

**Will the digital euro strengthen financial stability? Yes, within certain limits** with R. Bidder and T. Jackson; Deutsche Bundesbank Research Brief, 26 June 2024 - [Link](#)

**Measuring geopolitical risk: Perceptions matter** with Y. Bondarenko, V. Lewis and Y. Schüler; VoxEU Article, 5 June 2023 - [Link](#)

**Estimating Growth-at-Risk: Insights from a Structural Nonlinear Model**; SUERF Policy Brief, No 378, July 2022 - [Link](#)

**The Role of Contact Tracing in the Long Pandemic War** with L. Melosi; SUERF Policy Brief, No 242, Dec. 2021 - [Link](#)

**The reversal interest rate: A new motive for countercyclical macroprudential policy** with M. Darracq Pariès and C. Kok; VoxEU Article, 2 May 2021 - [Link](#)

**In Support of Monetary Policy: Using the Countercyclical Capital Buffer to Avoid a Reversal Interest Rate** with M. Darracq Pariès and C. Kok; SUERF Policy Brief, No 37, Nov. 2020 - [Link](#)

**Enhancing macroprudential space when interest rates are “low for long”** with M. Darracq Pariès and C. Kok; ECB Macroprudential Bulletin, Issue 11, 2020 - [Link](#)

**A macroprudential perspective on replenishing capital buffers** with K. Budnik, M. Darracq Pariès, C. Kok, J. Lang, M. Lo Duca, E. Rancoita, C. Rodriguez d’Acri and E. Ryan; ECB Financial Stability Review, Vol. 2, 2020 - [Link](#)

## Teaching

|                   |                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| 04/2026           | Deep Learning, neural networks, and generative modeling in macro (mini-course), European Commission, Instructor |
| 04/2025 - 08/2025 | Monetary Economics and Numerical Methods (Graduate), University of Erlangen-Nürnberg, Instructor                |
| 04/2023 - 08/2023 | Monetary Economics and Numerical Methods (Graduate), University of Erlangen-Nürnberg, Instructor                |

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 09/2019 - 09/2019 | Macro-Prudential Policy: A Quantitative Approach (Graduate), Florence School of Banking and Finance, TA for Enrique Mendoza |
| 11/2017 - 01/2018 | Macroeconomics I (Graduate), EUI, TA for Axelle Ferrière                                                                    |
| 10/2011 - 03/2014 | Statistics (Undergraduate) , University of Erlangen-Nürnberg, TA                                                            |

## Scholarships

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|-------------|---------------------------------------------------------------|
| 2021        | PhD Grant, European University Institute                      |
| 2016 - 2021 | PhD Scholarship, German Academic Exchange Service (DAAD)      |
| 2019        | U.S. Department Visiting Grant, European University Institute |

## Referee and Other Services

**Referee:** Review of Economic Studies, American Economic Journal: Macroeconomics, Journal of Monetary Economics, Journal of International Economics, European Economic Review, International Economic Review, Journal of Money, Credit and Banking, Journal of Economic Dynamics and Control, Oxford Bulletin of Economics and Statistics, Journal of International Money and Finance, Macroeconomic Dynamics, B.E. Journal of Macroeconomics, Economic Modelling

**Scientific committee member:** 4th Joint ESCB RCCC & NGFS ENR Workshop (2025), NGFS 4th Expert Network Research Workshop (2025), Deutsche Bundesbank's annual research conference (2024), ESCB Research Cluster 2 Annual Workshop (2021 & 2022)

## Presentations (incl. scheduled)

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| 2026 | European Commission, Oesterreichische Nationalbank, Swiss National Bank, CEBRA 2026 Annual Meeting, 2026 IJCB Conference, TU Wien,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2025 | Banque de France AI Methods Conference 2025, Inflation: Drivers and Dynamics 2025 Conference by ECB and Cleveland Fed, CEBRA 2025 Annual Meeting, ECB Conference on Financial Stability and Macroprudential Policy 2025, 3rd annual Non-Linearities in Macro Workshop by Bank of England, Queen Mary University of London & Lancaster University, Central Bank of Morocco, 2nd QRFE Workshop on Quantitative Finance, BSP-BIS Workshop on Macroeconomic Models, Joint Research Centre (JRC) Workshop on Energy Transition and Economic Decarbonisation, 4th Joint ESCB RCCC & NGFS ENR Workshop, ECB Working Group on Econometric Modelling, European Central Bank DG-Economics, Bank for International Settlements, University of St. Gallen |
| 2024 | University of Copenhagen, University of Zurich, Danmarks Nationalbank, ECB Banking Supervision Research Conference, KIT Quant Macro Day, European Commission Annual Research Conference (ARC) 2024, SNB Annual Research Conference, 5th Mannheim Workshop in Quantitative Macroeconomics, Bank for International Settlements, Eurosystem workshop: Digital euro & its holding limit, NGFS Research Webinar, ESCB Research Cluster on Monetary Economics, Czech National Bank Workshop on Financial Stability and Macroprudential Policy, Green Finance Research Advances 2024, Bank for International Settlements, Bank of Estonia                                                                                                            |
| 2023 | NBER Summer Institute, University of Bonn, Bank of Finland, Bank of Lithuania, Bank of Estonia, IWH Halle, IAAE 2023, 26th Theories and Methods in Macro (T2M), EEA-ESEM 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| 2022 | Banque de France, Schumpeter-BSE-Seminar (Berlin), ECB Working Group on Econometric Modelling - “Macro-at-Risk” Group, Ghent University Workshop on Empirical Macroeconomics, CEBRA 2022 Annual Meeting, 4th EUI Alumni Conference in Economics, Conference on advanced analytics: new methods and applications for macroeconomic policy (jointly organized by ECB, BoE and DAFM at King’s College London)                                                                                                           |
| 2021 | 8th Conference on New Developments in Business Cycle Analysis by Danmarks Nationalbank, Bundesbank, Norges Bank, ESCB Research Cluster 1 and 2 (discussant), SNB Research Conference 2021, IAAE 2021 Annual Conference, SEC 27th International Conference, QCGBF Annual Conference 2021, Federal Reserve Bank of Chicago, Boston College, Tartu University, Bank of Canada, University of Konstanz, Bank of England, Deutsche Bundesbank, University of Tübingen, Banque de France, KU Leuven, Federal Reserve Board |
| 2020 | Norges Bank, European Central Bank DG-Research, De Nederlandsche Bank 23rd Annual Research Conference, CEPR and Bank of Finland Joint Conference on Monetary Policy Tools, VfS Annual Conference 2020, 28th Annual Symposium of the Society for Nonlinear Dynamics and Econometrics, 4rd Annual Workshop of ESCB Research Cluster 3 (discussant), European Central Bank DG-MF, NBER SI 2020 Monetary Economics (co-author presented), Danmarks Nationalbank                                                          |
| 2019 | Northwestern University, Bank of Estonia Christmas Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2018 | Deutsche Bundesbank, Bank of Estonia Christmas Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                               |